Jillanne Youssef

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EDUCATION

University of Waterloo, BASc in Mechanical Engineering *Expected April* 2028

SKILLS

Mechanical Design: CAD (NX, CATIA V6, SolidWorks), DFMA (Injection Molding, Die-Casting, CNC-Machining)
Engineering Drawings: GD&T, Datum Reference Frames, Manufacturing & Inspection Drawings, Fits and Clearances
Analysis: Tolerance Analysis, FEA (Ansys Mechanical), CFD (Ansys Fluent), RCFA/FMEA, Mold Flow Simulation
Manufacturing & Prototyping: Machining (Lathe, Mill), 3D Printing (SLA, SLS, FDM), Waterjet & Laser Cutting
Electrical & Software: PCB Design (KiCad, Altium), Embedded (ESP32, Arduino, Raspberry Pi) C++, Python, MATLAB

EXPERIENCE

Google, Product Design Engineering Intern - Pixel Hardware *May* 2026 – *Aug* 2026 | *Mountain View, California*

- Owned the design and validation of the Pixel Watch screen geometry for precision-molded glass, using prototype testing and supplier DFM to eliminate optical distortion while meeting industrial design intent.
- Led design and DFM of Pixel Phone nano-molded antenna splits, eliminating leak-causing defects and improving bonding through localized leak testing while preserving industrial design intent and securing supplier approval.
- Saved $15K in vendor testing by developing a production-representative NMT test coupon and fixture that recreated a Pixel Watch titanium-resin interface, enabling seal-failure root-cause testing without complete watch builds.
- Developed an installed-condition drop test to validate dynamic FEA of a preloaded glass-filled plastic watch baffle, avoiding destruction of 15 low-volume prototypes per test cycle and saving $20K+ in prototype costs.
- Engineered a universal strain-test jig for resin dogbone specimens, using a mathematically derived elliptical profile and tool-free clamping to standardize 2% strain testing across 4 global suppliers and eliminate supplier-specific tooling.

Level Home, Product Design Engineering Intern *Jan* 2026 – *Apr* 2026 | *Redwood City, California*

- Proposed, prototyped, and secured approval for a new lock-sensing architecture, replacing accelerometers and FPCAs with integrated magnetometers and insert-molded magnets to reduce unit cost by $12.
- Led DFM for 20+ injection-molded and die-cast components, redesigning gates and parting lines through supplier tooling reviews to eliminate cosmetic defects and cut tooling costs 8% while preserving industrial design intent.
- Root-caused recurring lock-bolt binding through datum-based tolerance stack-ups and cycle testing, driving mechanical design revisions that reduced actuation-force variability 25% and improved mechanism pass rates 15%.
- Designed and built an automated solenoid test fixture and PCBA for keypad validation, uncovering 14 reliability issues across 2M+ cycles and collaborating with EE and firmware to achieve 100% EVT functional success.

Tesla, Product Design Engineering Intern - Lighting & Switches *Sep* 2025 – *Dec* 2025 | *Fremont, California*

- Led the design of a new program interior cabin light, using CATIA V6 for solid and surface modeling of Class A surfaces for injection molding, optimized for cost, ease of assembly, and user experience.
- Designed a 2-layer mixed-signal PCB from scratch in KiCad, specified and selected all components, hand-soldered and assembled the board, and optimized the design for low cost, DFM, and fabrication.
- Proposed and built an RGBW controller for internal company use, replacing off-the-shelf controllers by integrating a PCB, SLA-printed enclosure, and ESP32, streamlining design studio process by increasing LEDs per controller by 6x.

Tesla, Mechanical Design Engineering Intern - Cell Equipment *Jan* 2025 – *Apr* 2025 | *Palo Alto, California*

- Designed a high-temp magnetic clamping assembly for cell lamination, achieving 0.1 mm clearance through tolerance stackups and vibration-resistant design, boosting yield by 15% and cutting setup time from 30 to 5 seconds.
- Led mechanical, electrical, and controls design of a modular foil stretching machine with load-cell-based tension control, reducing foil wrinkling and creasing failures by 50%.
- Designed and built an electrical control box integrating Arduino, load cell, amplifier, and signal conditioner for closed-loop pneumatic control of nip rollers, maintaining roller nip force within ±2% of target under dynamic loading.